

Transverse pure gauge fields in Yang-Mills theory

F. A. Schaposnik

Laboratoire de Physique Théorique et Hautes Energies Orsay, France*

J. E. Solomin

Laboratoire de Géométrie et Analyse Complexe Université de Paris VI, U.E.R. 47

(Received 13 April 1978)

For an SU(2) Yang-Mills theory, we discuss some properties of gauge transformations leading to ambiguities recently observed in the Coulomb gauge.

Recently, Gribov¹ observed that, for an SU(2) Yang-Mills gauge theory, the transversality condition

$$\partial_i A^i = 0 \quad (1)$$

does not actually fix the gauge completely. In particular, for a vanishing field $F_{\mu\nu} = 0$, condition (1) can be written in the form

$$\partial_i (g^{-1} \partial^i g) = 0, \quad (2)$$

where $g(x)$ is a gauge-group element that generates a pure gauge potential

$$A_\mu = g^{-1} \partial_\mu g. \quad (3)$$

Gribov has shown that Eq. (2) has nontrivial solutions if one makes a particular ansatz for $g(x)$. Using the same ansatz, Jackiw, Muzinich, and Rebbi² clarified the types of ambiguities that occur in the Coulomb gauge showing that they are a necessity if one wants to accommodate field configurations with nonzero Pontryagin number. In this note we will discuss the relation between these results and the particular ansatz used for g . In order to do that, we begin by considering the quantity³

$$\varphi = -\frac{1}{24\pi^2} \int d^3x \epsilon_{ijk} \text{Tr}(\mathbf{A}_i \mathbf{A}_j \mathbf{A}_k) \quad (4)$$

associated with the topological properties of the solutions. The solution exhibited by Gribov corresponds to $|\varphi| = \frac{1}{2}$ (of course, for the trivial solution, $A_i = 0$ and $\varphi = 0$).

We will first show that, working at fixed time, under a simple set of assumptions, no nontrivial solution with $|\varphi| \neq \frac{1}{2}$ exists. Moreover, the $|\varphi| = \frac{1}{2}$ solution corresponds to the Gribov ansatz which turns out to be the only possible one.

Using this result we will easily demonstrate that under those assumptions any regular Euclidean configuration $A_\mu(\vec{x}, x_4)$ that connects vacuum states (3) at times $x_4 = \pm\infty$ can only have Pontryagin num-

ber $q = 0, \pm 1$ in accord with one of the results of Ref. 2. Equation (2) can be obtained as the extremum of the "action"

$$S = \int d^3x \partial_i g^\dagger \partial^i g \quad (5)$$

under the constraint $g^\dagger g = 1$.

Now, if we write for any $g \in \text{SU}(2)$

$$g = e^{i\vec{\alpha}(x) \cdot \vec{\sigma}} \quad (6)$$

(here $\vec{\sigma}$ are the Pauli matrices), then S reads

$$S = \int d^3x (\partial_i \alpha \partial^i \alpha + \sin^2 \alpha \partial_i \vec{n} \cdot \partial^i \vec{n}), \quad (7)$$

where $\vec{\alpha} = \alpha \vec{n}$, with the condition

$$|\vec{n}|^2 = 1. \quad (8)$$

This constraint can be incorporated in the "action" with the aid of a Lagrange multiplier λ ,

$$S = \int d^3x [\partial_i \alpha \partial^i \alpha + \sin^2 \alpha \partial_i \vec{n} \cdot \partial^i \vec{n} + \lambda(x)(\vec{n}^2 - 1)]. \quad (9)$$

Then, the "equations of motion" are

$$\partial_i \partial^i \alpha = \sin \alpha \cos \alpha \partial_i \vec{n} \cdot \partial^i \vec{n}, \quad (10a)$$

$$\partial_i (\sin^2 \alpha \partial^i \vec{n}) = \lambda \vec{n}, \quad (10b)$$

$$\vec{n}^2 = 1. \quad (10c)$$

We shall look for solutions of Eqs. (10) satisfying the following assumptions:

(i) separability into radial (r) and angular (θ, ϕ) variables in the form

$$\vec{\alpha} = \alpha(r) \vec{n}(\theta, \phi), \quad (11)$$

where these functions are *a priori* unknown [we do not assume the specific ansatz of Gribov, neither do we suppose $\vec{n}(\theta, \phi)$ to be spherically symmetric].

(ii) analyticity in the variables and single-valued-

ness. (As has been stressed by Chang and Chang,⁴ in Yang-Mills theories, multivalued gauge transformations are in principle admissible. This kind of transformation, excluded in the present note, is under study.)

Assumption (ii) allows us to write Eq. (10b) in the form

$$-\frac{\sin^2 \alpha}{r^2} L^2 \vec{n} = \lambda \vec{n}, \quad (12)$$

where L^2 is the orbital angular momentum operator

$$L^2 = -\frac{1}{\sin \theta} \partial_\theta (\sin \theta \partial_\theta) - \frac{1}{\sin^2 \theta} \partial_\phi^2. \quad (13)$$

Taking the scalar product of Eq. (12) by $\vec{n}$ yields

$$\vec{n} \cdot L^2 \vec{n} = \frac{-\lambda r^2}{\sin^2 \alpha}, \quad (14)$$

since $\vec{n}^2 = 1$, $\vec{n} \cdot \partial_i \vec{n} = 0$, and

$$\vec{n} \cdot L^2 \vec{n} = -r^2 \partial_i \vec{n} \cdot \partial^i \vec{n}. \quad (15)$$

Then, using Eqs. (14) and (15) we can write Eq. (10a) in the form

$$\frac{1}{r^2} \tan \alpha \frac{d}{dr} \left(r^2 \frac{d\alpha}{dr} \right) = \lambda. \quad (16)$$

Hence, $\lambda = \lambda(r)$ and

$$\vec{n} \cdot L^2 \vec{n} = \frac{-r^2 \lambda(r)}{\sin^2 \alpha(r)} = C, \quad (17)$$

so that we obtain from expression (12) the equation

$$L^2 \vec{n} = C \vec{n}. \quad (18)$$

According to the hypothesis of analyticity and single-valuedness on the sphere, C must be of the form $l(l+1)$, where l is a non-negative integer. Therefore, the "equations of motion" are now separated into radial and orbital parts

$$\frac{d}{dr} \left(r^2 \frac{d\alpha}{dr} \right) = l(l+1) \sin \alpha \cos \alpha, \quad (19)$$

$$L^2 \vec{n} = l(l+1) \vec{n}. \quad (20)$$

Equation (19) becomes a Gribov equation in the $l=1$ case. Moreover, it is easy to show that $l=0, 1$ are the only allowed cases. This is a direct consequence of the following general theorem:

Theorem: Let there be N real analytic functions on the sphere $n^a(\theta, \phi)$, satisfying

$$(a) \quad n^a n^a = 1,$$

$$(b) \quad L^2 n^a = l(l+1) n^a.$$

Then, the only allowed values of l are such that

$$2l+1 < N.$$

For a proof of this theorem, see Ref. 5.

Then, in the $N=3$ case, we have only two possibilities:

sibilities:

(1) $l=0$, which corresponds to a constant α (in this case, $A_i = g^{-1} \partial_i g = 0$),

(2) $l=1$, which reproduces the Gribov ansatz by choosing the axis in isospin space in a convenient fashion. Indeed, from assumption (ii) and Eq. (20), n^a can be written in the form

$$n^a = \sum_{m=-1,0,1} C_m^a Y_1^m(\theta, \phi). \quad (21)$$

Then, we have

$$\begin{aligned} n &= (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) \\ &= \left(\frac{4\pi}{3} \right)^{1/2} \left(\frac{-Y_1^1 + Y_1^{-1}}{\sqrt{2}}, \frac{-Y_1^1 - Y_1^{-1}}{\sqrt{2}i}, Y_1^0 \right), \end{aligned} \quad (22)$$

and for $\alpha(r)$, the equation discussed by Gribov with a solution satisfying

$$\alpha(0) = 0, \quad \alpha(r) \underset{r \rightarrow \infty}{=} \pm \pi/2,$$

which gives for A_i [Eq. (3)]

$$A_i \underset{r \rightarrow \infty}{\sim} \frac{\vec{\sigma} \times \vec{n}}{r} \Big|_i. \quad (22')$$

We have then proved the nonexistence of solutions with $|\varphi| \neq 0, \frac{1}{2}$ under conditions (i) and (ii); also, we showed that the only nontrivial one corresponds to the Gribov ansatz. We can now interpret these results in light of the analysis developed in Ref. 2. Since we are working at a fixed time, we can consider the mapping

$$g(x) = R^3 \rightarrow S^3 \quad (23)$$

between Euclidean space R^3 and the group manifold S^3 . Now, the integral [Eq. (4)] defining φ can be rewritten in the form

$$\varphi = \frac{-1}{2\pi^2} \int_{g(R^3)} dS, \quad (24)$$

where dS is the surface element on the S^3 sphere and $g(R^3)$ is the image of R^3 on S^3 by g .

For nontrivial solutions of Eqs. (2), $g(R^3)$ must have a nonvanishing boundary. [Otherwise, g will be a constant, from statement A of Ref. 3. Note that only in this case can g be extended to the compact $R^3 \cup (\infty) = S^3$ and then φ has a topological meaning.] Since one requires continuity of the mapping [assumption (ii)], such a boundary is homeomorphic to S^2 . Then, at spatial infinity we have the mapping

$$g(\infty, \theta, \phi) : S^2 \rightarrow S^2. \quad (25)$$

The topology of this mapping is labeled by an integer, $m[\pi_2(S^2) = Z]$. On the other hand, φ does not

characterize the topology of the mapping given by Eq. (23) since, R^3 being the domain, this mapping can be deformed continuously to a constant. Under our assumptions, the mapping $g(\infty, \theta, \phi)$ [Eq. (25)] is topologically equivalent to

$$n(\theta, \mathcal{L}) : \frac{\vec{x}}{r} \rightarrow n(\theta, \phi) \quad (26)$$

[since we are interested in the case $\alpha(r) \rightarrow_{r \rightarrow \infty} \pm \pi/2$].

But we found that the only nontrivial $\vec{n}$ can be written as $\vec{n} = \vec{x}/r$. Then, the integer m introduced above can only take the values

$$m = \pm 1 \quad (27)$$

[the sign depends on the choice of the generator of $\pi_2(S^2)$]. Then, from statement C of Ref. 3 we see that, when separability is assumed, $|\varphi| > 1$ is excluded (since this possibility corresponds to $|m| \geq 2$). In fact, the only allowed n corresponds to $\varphi = \pm \frac{1}{2}$. We finally point out that Eq. (20) excludes the obvious candidate for an $m \geq 2$ solution

$$\vec{n} = \begin{pmatrix} \sin\theta \cos m\phi \\ \sin\theta \sin m\phi \\ \cos\theta \end{pmatrix}. \quad (28)$$

Instead of ansatz (28), one must try more complicated and nonseparable possibilities. For example, the axially symmetric ansatz

$$\vec{a} = \begin{pmatrix} \alpha(\rho, Z) \cos m\phi \\ \alpha(\rho, Z) \sin m\phi \\ \beta(\rho, Z) \end{pmatrix}, \quad (29)$$

a possibility we do not exclude.

As we stated at the beginning, our result was obtained working at a fixed time. However, we can use it to study quantum tunnelling between possible vacuums. From the analysis of Ref. 3, we know that in the Coulomb gauge a regular Euclidean configuration $F_{\mu\nu}(\vec{X}, X_4)$ with finite action which starts from a vacuum with a given m , must have $|q| = |m|$ or $q = 0$. Then, if at $X_4 = \pm\infty$ we have vacuum states of the form (3), obtained under assumption (i) and (ii), any regular Euclidean configuration $A_\mu(\vec{X}, X_4)$ connecting them must have $q = 0, \pm 1$.

As we stated in the beginning, this result is in accord with the analysis of Jackiw, Muzinich, and Rebbi.² These authors considered spherically symmetric configurations and studied gauge transverse potentials with $\vec{n} = \vec{X}/r$. They conclude that if one excludes transverse potentials with a long-range $O(1/r)$ tail, discontinuities are necessary in order to have $q \neq 0$. If the $O(1/r)$ behavior is admitted, as in the case of the potential given by Eqs. (3)-(22'), then only $|q| \leq 1$ configurations can be accommodated without discontinuities.

ACKNOWLEDGMENTS

One of the authors (J.E.S.) would like to thank Professor Dolbeault for his hospitality at the Laboratoire de Géométrie et Analyse complexe, Université de Paris VI. The work of J.E.S. was supported by CONICET, Argentine.

*Laboratoire associé au Centre National de la Recherche Scientifique. Postal address: Bât. 211, Université Paris-Sud, 91405 Orsay, France.

¹V. N. Gribov, lecture at the 12th Winter School, Leningrad, Nuclear Physics Institute report, 1977 (unpublished).

²R. Jackiw, I. Muzinich, and C. Rebbi, Phys. Rev. D **17**,

1576 (1978).

³M. Ademollo, E. Napolitano, and S. Scuitto, Nucl. Phys. B **134**, 477 (1978).

⁴L. N. Chang and N. P. Chang, Phys. Lett. **72B**, 341 (1978).

⁵E. Cremmer, F. Schaposnik, and J. Scherk, Phys. Lett. **65B**, 78 (1976).